Dear Hiring Manager,

I have over seven years of experience in mechanical design, where I developed strong skills in problem-solving and project management. I'm now looking to transition into the field of artificial intelligence.

As an R&D engineer at Tam-Taş Packaging Machinery, a manufacturer in packaging machinery industry, I designed semi/full-automatic filling and packaging lines including vertical and rotational packaging, conveyor systems and side equipment. I became familiar with AI and computer vision through Optical Character Recognition (OCR) cameras used in these filling lines.

I worked as a Senior R&D Engineer at Transvaro Electron for the past two and a half years, where I was involved in the design and development of advanced electro-optical systems, including both clip-on and stand-alone devices suitable for day and night use. My responsibilities included designing systems based on thermal imaging and holographic technologies. These experiences have strengthened my interest in vision technologies and motivated me to pursue a new direction in my career within this area.

I resigned from my job in October 2023 to devote more time to my graduate courses and make a career change a serious goal. I am a full-time graduate student pursuing a master's degree in Electrical and Electronics Engineering at Koç University. I successfully completed my coursework in the Fall term of 2024, which included classes such as Introduction to Machine Learning, Computer Vision with Deep Learning, Medical Image Analysis with Machine Learning, Computational Imaging, Digital Image Processing, and Photonics and Lasers. I am currently working on my thesis, which focuses on spherical and chromatic aberration correction in electro-optical systems using deep learning methods.

Throughout my studies and research, I have worked with deep learning architectures like CNNs, RNNs, GANs, diffusion and ViTs, alongside optimization techniques such as Adam, SGD and learning rate scheduling and model development, including fine-tuning, custom architectures and training strategies for image re-generation, object detection, object tracking, and segmentation tasks. I am familiar with different kinds of image processing techniques, such as FIR filters, multiscale image representations, image filtering, image de-blurring, image denoising (Gaussian, Median, Bilateral filters), single-image super-resolution and optical flow. I can proficiently implement these methods using PyTorch framework and OpenCV library.

Should you consider my qualifications align with the requirements of the role, I would like to further elaborate on how my skills and experience can contribute to your organization's engineering-related efforts and express my enthusiasm for the prospect of engaging in a discussion with you regarding the position.

Sincerely,

Arda Uludağ

Arda Uludağ

Istanbul, Turkey | (+90) 535 495 79 35

LinkedIn | GitHub | E-mail 1 | E-mail 2

SKILLS

Python | Pytorch | Numpy | Scikit-learn | Matplotlib | Machine Learning | Deep Learning | Deep Learning based Computer Vision | CNN Architecture | Generative Adversarial Network Architecture | Transformer Architecture | Diffusion Architecture | RNN Architecture | Data Handling | Data Preprocessing

WORK EXPERIENCE

Transvaro Elektron Jun 2021 - Oct 2023

Senior Research and Development Engineer

- Conducted cutting-edge research in hologram, photonics, and thermal technologies.
- Designed and developed innovative solutions and prototypes in electro-optical systems.
- Performed simulations and experiments to test and validate new technologies.
- Managed project timelines, resources, and deliverables to ensure successful project outcomes.

Miuul Feb 2022 - Jun 2022

Data Science & Machine Learning Bootcamp Participant

- Conducted focused studies on feature engineering, machine learning, MS SQL, and Natural Language Processing (NLP) to enhance data processing and model performance.
- Prepared data sets for machine learning models using advanced feature engineering techniques, including handling outliers and missing values, encoding and scaling features, and feature extraction to improve model accuracy.
- Tackled regression and classification problems utilizing machine learning algorithms such as Linear Regression, Logistic Regression, KNN (K-Nearest Neighbors), CART (Classification and Regression Trees), Random Forests, including XGBoost, LightGBM, and CatBoost.
- Developed and maintained a comprehensive Machine Learning Pipeline, including exploratory data analysis (EDA), data pre-processing, creating and validating base models, automated hyperparameter optimization, Stacking and Ensemble Learning techniques, and prediction and deployment of final models.

Tam-Taş Packaging Machinery Apr 2017 - May 2021

Mechanical Design Engineer

- Designed semi and full-automatic filling lines, including vertical and rotational packaging, and conveyor systems.
- Expertise in mechanical drafting, documentation, and machining.
- Managed project/product lifecycle: scheduling, planning, design, and testing.

PROJECTS

High-Quality Image Regeneration from Deblurred Images Using GAN

A Comparison of Traditional Machine Learning and Deep Learning Approaches for Chest X-Ray Analysis for Pneumonia Detection

Scene Segmentation in Autonomous Driving System using Label-Efficient Visual Abstractions

Customer Churn Prediction

Customer Segmentation with Unsupervised Learning

CERTIFICATIONS

Python Programming Language Feb 2024
C & System Programmers Association

Coursera Aug 2022
Advanced Learning Algorithms Certification

Coursera Aug 2022
CNNs in Tensorflow Certification

Coursera July 2022
Supervised Machine Learning: Regression and Classification Certification

EDUCATION

MSc in Electrical and Electronics Engineering Feb 2023 - Dec 2025
Specialization in Artificial Intelligence
Koç University

Bachelor's Degree - Mechanical Engineering Sep 2010 - June 2014
Manisa Celal Bayar University